Multiplying and Dividing Fractions

Name _____ Date _____ No calculator Version 1

KEY WORDS numerator · denominator · product · reciprocal · simplify · mixed number

A A fraction of an amount ●○○ ANSWER ON THE SHEET

Remember: Divide by the bottom, then multiply by the top: $\frac{3}{4}$ of 20 is $20 \div 4 \times 3 = 15$.

1. Work out.

a) $\frac{1}{4}$ of 28 = _____ **b)** $\frac{2}{3}$ of 18 = _____ **c)** $\frac{3}{5}$ of 35 = _____ **d)** $\frac{5}{6}$ of 42 = _____

e) $\frac{3}{8}$ of 56 = _____ **f)** $\frac{7}{10}$ of 90 = _____ **g)** $\frac{4}{9}$ of 63 = _____ **h)** $\frac{2}{7}$ of 84 = _____

i) $\frac{3}{4}$ of 120 m = _____ **j)** $\frac{5}{8}$ of 240 g = _____ **k)** $\frac{2}{5}$ of 65 km = _____ **l)** $\frac{11}{12}$ of 60 min = _____

B A fraction times a whole number ●○○ ANSWER ON THE SHEET

Remember: Multiply the top by the whole number. Write an improper fraction as a mixed number.

2. Work out. Simplify your answer where you can.

a) $\frac{1}{3} \times 2$ = _____ **b)** $\frac{2}{5} \times 2$ = _____ **c)** $\frac{3}{4} \times 3$ = _____ **d)** $\frac{2}{7} \times 5$ = _____

e) $\frac{5}{6} \times 4$ = _____ **f)** $\frac{3}{8} \times 6$ = _____ **g)** $\frac{4}{9} \times 3$ = _____ **h)** $\frac{7}{10} \times 5$ = _____

C Multiplying fractions ●●○ WORK IN YOUR BOOK

Remember: Multiply the tops together and multiply the bottoms together. Then simplify.

3. Work out. Simplify your answer where you can.

a) $\frac{1}{2} \times \frac{1}{3}$ **b)** $\frac{2}{3} \times \frac{1}{5}$ **c)** $\frac{3}{4} \times \frac{1}{2}$

d) $\frac{2}{5} \times \frac{3}{7}$ **e)** $\frac{3}{4} \times \frac{2}{3}$ **f)** $\frac{5}{6} \times \frac{3}{10}$

g) $\frac{4}{9} \times \frac{3}{8}$ **h)** $\frac{7}{8} \times \frac{4}{5}$ **i)** $\frac{5}{12} \times \frac{3}{5}$

j) $\frac{9}{10} \times \frac{5}{6}$ **k)** $\frac{7}{3} \times \frac{6}{5}$ **l)** $\frac{8}{3} \times \frac{9}{4}$

© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.

D Reciprocals ●●○ — ANSWER ON THE SHEET

> **Remember:** To find the reciprocal, turn the fraction upside down. A number times its reciprocal is 1.

4. Write the reciprocal of each number.

a) $\frac{3}{4}$ _____ **b)** $\frac{2}{5}$ _____ **c)** 5 _____ **d)** $\frac{1}{6}$ _____

e) $\frac{7}{3}$ _____ **f)** 12 _____ **g)** $2\frac{1}{2}$ _____ **h)** $1\frac{2}{3}$ _____

E Dividing fractions ●●● — WORK IN YOUR BOOK

> **Remember:** To divide by a fraction, multiply by its reciprocal: $\frac{2}{3} \div \frac{4}{5} = \frac{2}{3} \times \frac{5}{4}$.

5. Work out. Simplify your answer where you can.

a) $\frac{1}{2} \div \frac{1}{4}$ **b)** $\frac{2}{3} \div \frac{1}{3}$ **c)** $\frac{3}{4} \div \frac{1}{2}$

d) $\frac{2}{5} \div \frac{3}{10}$ **e)** $\frac{5}{6} \div \frac{2}{3}$ **f)** $\frac{3}{8} \div \frac{3}{4}$

g) $\frac{4}{7} \div \frac{2}{5}$ **h)** $\frac{7}{9} \div \frac{2}{3}$ **i)** $\frac{5}{8} \div \frac{15}{16}$

j) $\frac{3}{10} \div \frac{6}{5}$ **k)** $\frac{9}{4} \div \frac{3}{8}$ **l)** $\frac{7}{12} \div \frac{14}{3}$

F Problems ●●● — WORK IN YOUR BOOK

6. One batch of muffins needs $\frac{3}{4}$ of a cup of milk. How much milk is needed for 6 batches?

7. A 6 m rope is cut into pieces that are each $\frac{3}{8}$ m long. How many pieces are there?

8. A tank holds 45 litres when it is full. It is $\frac{3}{5}$ full. How many litres are in the tank?

9. There are 24 students in a class. $\frac{2}{3}$ of them walk to school, and $\frac{1}{4}$ of the walkers live in the same street. How many students is that?

★ Extension — FOR EARLY FINISHERS

10. Change each mixed number to an improper fraction first.

a) $1\frac{1}{2} \times 2\frac{2}{3}$ **b)** $3\frac{3}{4} \div 1\frac{1}{4}$

© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.

Multiplying and Dividing Fractions

Name _____ Date _____ No calculator Version 2

KEY WORDS numerator · denominator · product · reciprocal · simplify · mixed number

A A fraction of an amount ●○○ ANSWER ON THE SHEET

Remember: Divide by the bottom, then multiply by the top: $\frac{3}{4}$ of 20 is $20 \div 4 \times 3 = 15$.

1. Work out.

a) $\frac{1}{5}$ of 35 = _____ **b)** $\frac{2}{3}$ of 24 = _____ **c)** $\frac{3}{4}$ of 36 = _____ **d)** $\frac{5}{6}$ of 54 = _____

e) $\frac{3}{8}$ of 72 = _____ **f)** $\frac{7}{10}$ of 60 = _____ **g)** $\frac{4}{9}$ of 81 = _____ **h)** $\frac{2}{7}$ of 63 = _____

i) $\frac{3}{5}$ of 150 m = _____ **j)** $\frac{5}{8}$ of 320 g = _____ **k)** $\frac{2}{3}$ of 75 km = _____ **l)** $\frac{7}{12}$ of 60 min = _____

B A fraction times a whole number ●○○ ANSWER ON THE SHEET

Remember: Multiply the top by the whole number. Write an improper fraction as a mixed number.

2. Work out. Simplify your answer where you can.

a) $\frac{1}{4} \times 3$ = _____ **b)** $\frac{2}{7} \times 3$ = _____ **c)** $\frac{2}{3} \times 4$ = _____ **d)** $\frac{3}{5} \times 4$ = _____

e) $\frac{5}{8} \times 4$ = _____ **f)** $\frac{5}{6} \times 3$ = _____ **g)** $\frac{2}{9} \times 6$ = _____ **h)** $\frac{3}{10} \times 5$ = _____

C Multiplying fractions ●●○ WORK IN YOUR BOOK

Remember: Multiply the tops together and multiply the bottoms together. Then simplify.

3. Work out. Simplify your answer where you can.

a) $\frac{1}{2} \times \frac{1}{5}$ **b)** $\frac{3}{4} \times \frac{1}{3}$ **c)** $\frac{2}{3} \times \frac{1}{4}$

d) $\frac{3}{5} \times \frac{2}{7}$ **e)** $\frac{2}{3} \times \frac{3}{8}$ **f)** $\frac{5}{6} \times \frac{2}{5}$

g) $\frac{3}{10} \times \frac{5}{9}$ **h)** $\frac{5}{8} \times \frac{4}{7}$ **i)** $\frac{7}{12} \times \frac{3}{7}$

j) $\frac{4}{5} \times \frac{5}{6}$ **k)** $\frac{5}{2} \times \frac{4}{3}$ **l)** $\frac{9}{4} \times \frac{8}{3}$

© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.

D Reciprocals ●●○ — ANSWER ON THE SHEET

> **Remember:** To find the reciprocal, turn the fraction upside down. A number times its reciprocal is 1.

4. Write the reciprocal of each number.

a) $\frac{2}{3}$ _____ **b)** $\frac{3}{8}$ _____ **c)** 7 _____ **d)** $\frac{1}{9}$ _____

e) $\frac{9}{4}$ _____ **f)** 15 _____ **g)** $3\frac{1}{2}$ _____ **h)** $1\frac{3}{4}$ _____

E Dividing fractions ●●● — WORK IN YOUR BOOK

> **Remember:** To divide by a fraction, multiply by its reciprocal: $\frac{2}{3} \div \frac{4}{5} = \frac{2}{3} \times \frac{5}{4}$.

5. Work out. Simplify your answer where you can.

a) $\frac{1}{2} \div \frac{1}{6}$ **b)** $\frac{3}{4} \div \frac{1}{4}$ **c)** $\frac{2}{3} \div \frac{1}{2}$

d) $\frac{3}{5} \div \frac{9}{10}$ **e)** $\frac{5}{6} \div \frac{1}{3}$ **f)** $\frac{5}{8} \div \frac{3}{4}$

g) $\frac{3}{7} \div \frac{2}{5}$ **h)** $\frac{8}{9} \div \frac{2}{3}$ **i)** $\frac{3}{8} \div \frac{9}{16}$

j) $\frac{7}{10} \div \frac{14}{5}$ **k)** $\frac{8}{3} \div \frac{4}{9}$ **l)** $\frac{5}{12} \div \frac{10}{3}$

F Problems ●●● — WORK IN YOUR BOOK

6. One batch of muffins needs $\frac{2}{3}$ of a cup of milk. How much milk is needed for 5 batches?

7. A 9 m rope is cut into pieces that are each $\frac{3}{4}$ m long. How many pieces are there?

8. A tank holds 56 litres when it is full. It is $\frac{5}{8}$ full. How many litres are in the tank?

9. There are 30 students in a class. $\frac{3}{5}$ of them walk to school, and $\frac{1}{3}$ of the walkers live in the same street. How many students is that?

★ Extension — FOR EARLY FINISHERS

10. Change each mixed number to an improper fraction first.

a) $2\frac{1}{2} \times 1\frac{3}{5}$ **b)** $4\frac{1}{2} \div 1\frac{1}{2}$

© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.